

# Multi-Loan Approval System (INR ₹) — Requirements & Data Spec

ML Supervised Classification Pipeline: Tkinter GUI, KNN & Decision Tree

## 1. TARGET KAGGLE DATASET SPECIFICATION

- **Dataset Source:** Kaggle — *Loan Approval Dataset* by **Arbaaz Tamboli**.
- **Problem Architecture:** Supervised Binary Classification (Predicting Approval: 1 / 0).
- **Indian Rupee (₹) Calibration:** Financial entries scaled to Indian retail banking ranges (Lakhs/Crores).
- **Credit Rating Benchmark:** Calibrated to Indian **CIBIL Score** standards (300 to 900).

## 2. DATASET SCHEMA & TKINTER FIELD MAPPING

| Dataset Feature       | Data Type            | Standard / Unit (₹ INR)                     | Tkinter GUI Component                        |
|-----------------------|----------------------|---------------------------------------------|----------------------------------------------|
| Loan_Purpose          | Categorical          | Home, Personal, Business, Study, Gold, Card | Dropdown: <b>Select Loan Type</b>            |
| Loan_Type             | Categorical          | Secured / Unsecured                         | Auto-selected / Dropdown toggle              |
| Annual_Income         | Numeric (Float)      | ₹ INR (₹3,00,000 to ₹50,00,000+)            | Entry Field: <b>Annual Income (₹)</b>        |
| Loan_Amount_Requested | Numeric (Float)      | ₹ INR (₹50,000 to ₹1,50,00,000)             | Entry Field: <b>Loan Amount (₹)</b>          |
| Loan_Term             | Numeric (Int)        | Tenure in Months (12 to 240 months)         | Entry Field / Spinbox: <b>Tenure</b>         |
| Credit_Score          | Numeric (Int)        | CIBIL Score (300 to 900)                    | Entry Field: <b>CIBIL Score</b>              |
| Monthly_Expenses      | Numeric (Float)      | ₹ INR (Monthly living expenses & EMIs)      | Entry Field: <b>Monthly Obligations (₹)</b>  |
| Outstanding_Debt      | Numeric (Float)      | ₹ INR (Current pending debt)                | Entry Field: <b>Existing Liabilities (₹)</b> |
| Loan_Approval_Status  | Categorical (Target) | Approved (1) / Rejected (0)                 | Dynamic Color Banner (Green / Red)           |

## 3. FUNCTIONAL & NON-FUNCTIONAL REQUIREMENTS

### Functional Requirements

- **Data Validation:** Tkinter inputs strictly validated (CIBIL between 300–900, positive INR amounts).
- **Pipeline Preprocessing:** One-Hot Encoding for categorical features; `StandardScaler` applied for numeric columns.
- **Dual-Model Support:** User can evaluate using KNN, Decision Tree, or execute side-by-side comparative inference.
- **Confidence Probability:** Display approval likelihood percentage via `predict_proba()`.
- **Model Serialization:** Offline models exported to `scaler.pkl`, `knn_model.pkl`, and `dt_model.pkl`.

### Non-Functional Requirements

- **Inference Latency:** GUI prediction returned within 200 ms upon clicking predict.
- **Distance Metric Integrity:** Compulsory feature normalization so rupee magnitudes do not skew KNN distance computations.
- **Reproducibility:** Fixed random state (`random_state=42`) across splitting, pruning, and folds.
- **Robust Error Handling:** Non-numeric or missing entries trigger clean GUI warning dialogs without crashing.

## 4. MODEL COMPARISON ARCHITECTURE (KNN VS. DECISION TREE)

| Evaluation Metric | K-Nearest Neighbors (KNN) | Decision Tree Classifier                              |
|-------------------|---------------------------|-------------------------------------------------------|
|                   |                           | Invariant — Unaffected by monetary scale differences. |

| Evaluation Metric         | K-Nearest Neighbors (KNN)                                        | Decision Tree Classifier                                                                        |
|---------------------------|------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Feature Scale Sensitivity | High — Requires StandardScaler transformation.                   |                                                                                                 |
| Decision Rationale        | Calculates distance vector to nearest Indian applicant records.  | Orthogonal IF-ELSE decision boundaries based on criteria cutoffs.                               |
| Inference Execution       | Computes distance across training vectors ( $\mathcal{O}(nd)$ ). | Instant traversal through conditional tree nodes ( $\mathcal{O}(\text{ext}\{\text{depth}\})$ ). |
| Hyperparameters           | n_neighbors, weights, metric (Euclidean/Manhattan).              | criterion, max_depth, min_samples_split, min_samples_leaf.                                      |